

# WireClaim

THE COMPLETE VERSION — the two-page write-up is [writeup.pdf](#)

## Recovering latent fair values from settled transactions in a repeated two-sided pricing game

Team **Bin busy** • QuantCo *Claim to Fame* (Track 1) • Munich Agentic Hackathon, 22–23 August 2026

**Abstract.** Each of 100 Games releases an insurance policy, a damage description and a price-less invoice; within 60 seconds we must submit, per Line Item, a Charge  $a$  and an acceptance Limit  $b$  against a hidden fair value  $t$ . We show (i) that the payoff table makes the Charge a *cliff* rather than a slope, so the optimal Charge sits strictly below the estimate; (ii) that  $t$  is *exactly recoverable* after settlement from the published transaction log, which turns every design question into a measurement; and (iii) that under that instrument the whole remaining gap to optimal play is estimation error, not strategy. We report a climb from last place to 5th, the second-best per-Game rate in the field over the last twenty Games, and the second-best risk-adjusted return.

### 1 The game, and the row everybody skips

Per ordered pair (issuer  $H$ , reviewer  $I$ ) and Line Item, with hidden fair value  $t$  and hidden cap  $c \geq \max(4t, 2000)$ , the reviewer’s payoff is

$$\pi_I = \begin{cases} -a & a \leq t, a \leq b \\ -1.5a & a \leq t, a > b \\ -\min(a, c) & a > t, a \leq b \\ 0 & a > t, a > b \end{cases} \quad (1)$$

while the issuer receives  $a$  whenever  $a \leq t$  — *whatever the reviewer does* — and  $\min(a, c)$  only if an Overcharge is accepted. So a fair Charge is owed by all  $N = 16$  opponents; an Overcharge is paid by the few whose Limit happens to admit it. Measured on Game 62: **16 payers versus 3**. The Charge is a **cliff, not a slope**: one rival charged 22% less than us on a single Line Item, was paid by 16/16, and took 136,075 — 87% of their whole Game.

### 2 Both decisions fall out of (1)

**The Charge.** With  $F$  the posterior CDF of  $t$  and  $q$  the fraction of opponents whose Limit admits  $a$ , income per opponent is  $a[1 - F(a) + F(a)q(a)]$ . Empirically  $q$  collapses across  $a = t$  (16/16 below, 17% at 1.0–1.3 $\times$ , 7% above), and a *mis-measured*  $q$  costs more than assuming  $q = 0$ , so we maximise the conservative objective. For  $t \sim \text{LogN}(\ln m, \sigma^2)$  and  $a = k m$ ,

$$R(k) \propto k \Phi\left(-\frac{\ln k}{\sigma}\right), \quad a = k(\sigma) m, \quad (2)$$

whose maximiser is *strictly below 1 at every precision* and falls as the estimate degrades (0.70 at  $\sigma=0.25$ , 0.60 at 0.45, 0.45 at 0.75). We ship  $k(\sigma) = \text{clamp}(0.85 - 0.45\sigma, 0.30, 0.80)$ .

**The Limit.** By (1) accepting costs  $a$ ; rejecting costs  $1.5a$  *only if the claim was fair*. Accepting is therefore right exactly when  $a < 1.5a P(t \geq a)$ , i.e.  $P(\text{fair}) > \frac{2}{3}$ , so the Limit is the **one-third quantile of the posterior** — derived, not tuned. Coverage needs no branch: with  $p = P(\text{covered})$  the posterior is a spike-and-slab,

$$P(t) = (1 - p) \delta_0 + p \text{LogN}(\ln m, \sigma^2), \quad (3)$$

so when  $p \leq \frac{2}{3}$  the bottom third *is* the atom at zero and  $b = 0$  emerges on its own. 40% of settled Line Items have  $t = 0$ , and paying on one is a pure loss.

### 3 The model reads; the engine prices

No model output is ever a price. Models emit *structured evidence* — a coverage probability with the policy

clause quoted verbatim, a price band, a quantity — and deterministic code turns evidence into the posterior and the posterior into  $(a, b)$ , combining channels by inverse-variance weighting in log space ( $w_i = \sigma_i^{-2}$ ,  $\sigma_{\text{memory}} = 0.43$  against  $\sigma_{\text{model}} = 0.60$ ).

The reason is auditability under repetition: this ran unattended 100 times, once every 12.6 minutes, in a 60-second window. **A prompt that emits a number cannot be swept, folded or replayed; a prompt that emits evidence can.** It also degrades rather than fails — **Games 82–89, eight consecutive 3 $\times$ -weighted Games, ran with the model endpoint returning 401 on every call and still banked +254,092.**

### 4 Why our counterfactuals are measurements

The published transaction log inverts. A *rejected* transaction carrying a non-zero amount was a wrongful rejection, which by row 2 of (1) occurs only when the Charge was fair; a rejection at zero occurs only when it was not. Hence

$$t \in \left[ \max_{\text{rej, amt} > 0} a_\alpha, \min_{\text{rej, amt} = 0} a_\alpha \right) \quad (4)$$

brackets  $t$  for *every Line Item ever played*. Over 52,224 settled rows the reconstruction reproduces every published net **to the cent**, so we can replay any hypothetical submission against the real Field with all sixteen opponents held fixed. The replay’s self-check asserts that our *actual* submission reproduces the authoritative net for every settled Game — a test, not a comment.

**The bar.** Positive on **all four folds** (odd, even, early, late), never merely in total, against a *measured* noise floor of 26,622 over 18 Games —  $\pm 6,275$  for one Game, which is why no single Game ever justified a change. Of twenty numbered hypotheses, **twelve were measured and rejected**. One shipped and was **reverted twenty minutes later**: it passed a real calibration table and all four folds, but a mechanism check showed it moved **4 Line Items in 573**. A constant can be right about its data and wrong about its mechanism.

### 5 What we removed, and what we rejected

We also *removed* a clamp. Until Game 66 the code enforced  $b \leq a$ ; since  $b$  is a lower quantile of the same posterior  $a$  is drawn from, this looks harmless and is not. Releasing it is worth +19,273 over 65 Games on all four folds and binds on 33% of items. Game 65 item 3 found it: our estimate was accurate to 1.4%, the clamp pinned  $b = a = 199.01$ , and thirteen *fair* Charges between 201

and 291 were all wrongfully rejected —  $-4,953$  on one Line Item.

A representative selection of the twelve rejections, each with the script that killed it: a global Charge multiplier below 1.0 (monotone loss,  $-118,049$  at  $\times 0.9$ ); the exact argmax of expected income (loses in **all 15 cells** of a two-parameter sweep); four separate attacks on the Limit (the derived  $\frac{1}{3}$  is the argmax); reading the policy’s exact EUR caps (the number is exact, the *row* is not identifiable,  $-5,905$ ); more ensemble draws (draw noise is 1.1% of the error, a third draw buys 0.2%); and copying the best rival’s  $a/t$  and  $b/t$  ratios, which applied to our own  $\hat{t}$  is  $-280,000$  to  $-380,000$ .

**Two traps, each of which cost a working session.** *Conditioning on the outcome:* bucketed by what items *turned out* to be worth, strictness pays 9.9:1 under 100 and loses above 500 — a compelling case for raising the Limit on expensive items. Bucketed by  $\hat{t}$ , the only split available at submission time, the gradient collapses: **of 23 items we believed were worth over 2,000, 14 were worth under half that.** *Censoring:* a Fair Value bracket with no upper bound is unbounded *because* nobody rightfully rejected, which correlates with the item being expensive — so every  $\sigma$  we quote is optimistic, and we say so.

## 6 What we would fix next, and why it is all one thing

Both numbers we submit are functions of the *same* estimate.  $a = k(\sigma)\hat{t}$  and  $b = Q_{1/3}$  of a posterior centred on  $\hat{t}$ . So a single estimation error is paid **twice, in opposite directions**, and this is the whole of our remaining gap:

- $\hat{t}$  **too low**  $\Rightarrow b$  too low  $\Rightarrow$  we wrongfully reject a *fair* claim  $\Rightarrow$  we pay **1.5a**. The half-charge goes to nobody. This is the largest single line in our costs, and note it fires *only* on claims that were fair — rejecting a genuine Overcharge is free.
- $\hat{t}$  **too high**  $\Rightarrow a > t \Rightarrow$  thirteen of sixteen opponents refuse it  $\Rightarrow$  we earn **nothing** on an item we were owed money for. Crossing  $t$  costs roughly  $5\times$  the income.

That is why  $a = b = t$  reaches 100.3% of optimal play while  $a = b = \hat{t}$  scores  $-50,140$ : the 2,498,118 between them is not strategy, and no decision rule reaches it. In priority order, what we would build with more time:

1. **Calibrate the band — worth more than any constant in the engine.** The model asserts a median  $\sigma$  of 0.375 against a realised RMSLE near 0.80: overconfident by  $2.1\times$ . Worse, the width carries *no signal* — split items by band width and the narrow third scores RMSLE 0.847 against the wide third’s 0.733, i.e. slightly *backwards*. So  $k(\sigma)$  multiplies a number that does not measure what it claims to. The line is right; its input is not.
2. **Fix the coverage read.** Policy clause 7.1.5 confines indemnity to the affected parts, *but* adds that where a room was wetted as a whole, the whole room is treated as affected. The model reads the first half only. Game 74 returned coverage 0.01–0.05 on 20 of 31 Line Items, 87% of that Game’s penalties. Perfect coverage is worth  $+1,173/\text{Game}$  on all four folds.
3. **Separate the genuinely large items from the phantoms.** Of 23 items we believed worth over 2,000, **14 were worth under half that.** An observable that

splits the 9 real ones would let us charge confidently where it is safe — today we cannot, so we charge low everywhere.

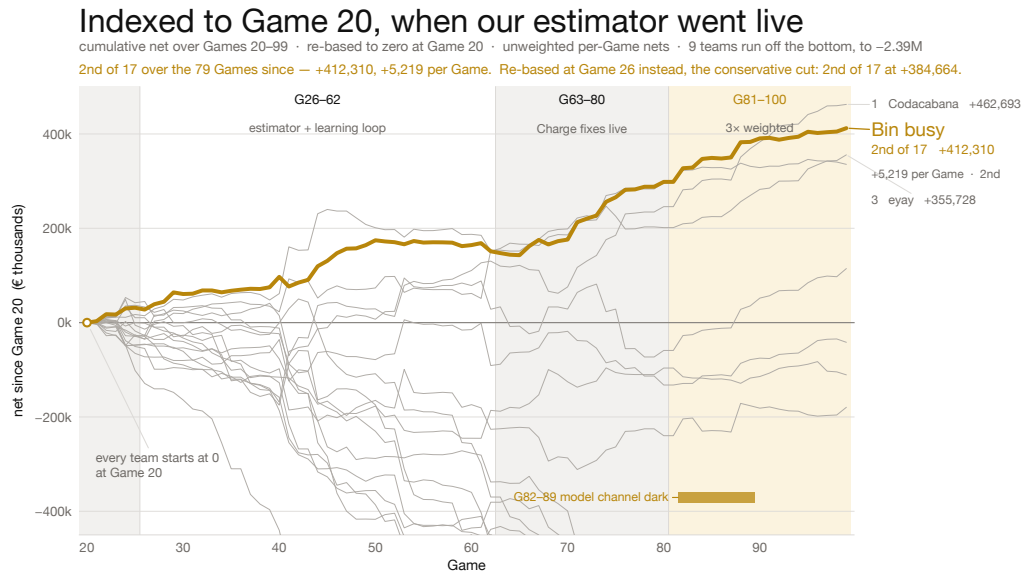
4. **Widen the memory channel.** A wording we have watched settle has a measured log error of **0.43** against the model’s realised 1.66–2.20, but memory only reaches about 22% of Line Items. Every point of coverage there is a direct reduction in  $\sigma$  — and by the first item above,  $\sigma$  is what sets both numbers.

## 7 Results

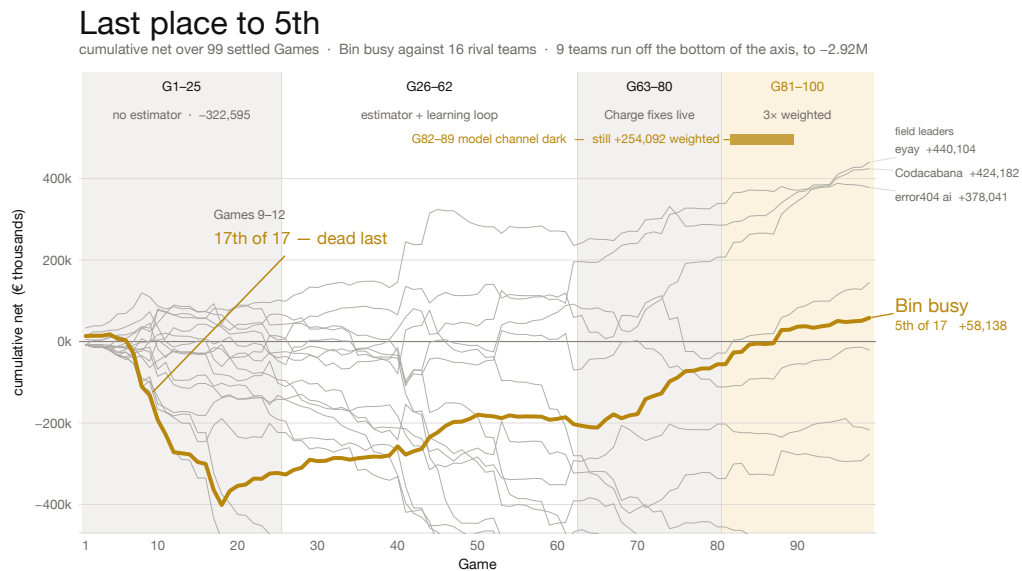
| window                                 | total                        | per Game  | rate rank     |
|----------------------------------------|------------------------------|-----------|---------------|
| G1–25 ( <i>estimator not working</i> ) | <b><math>-322,595</math></b> | $-12,904$ | —             |
| G26–94 ( <i>since</i> )                | $+362,531$                   | $+5,254$  | <b>3 / 17</b> |
| G75–94 ( <i>last twenty</i> )          | $+137,942$                   | $+6,897$  | <b>2 / 17</b> |
| season, weighted                       | $+231,298$                   | —         | <b>5 / 17</b> |

We were **17th of 17 — last — after nine Games**, and 5th from Game 76; the G1–25 deficit exceeds our entire season net, so the deficit *is* the learning curve. The result we would defend is the variance: **worst Game of the last twenty just**  $-3,941$ , and 3 losing in the last thirty, second fewest, for a mean-to- $\sigma$  of **0.75 — second in the field.**

Season total asks how much money we ended up with, which for us is largely a question about Games 1–19: our cumulative net at Game 20 was  $-354,171$ , and the whole gap to the leaders is that hole rather than anything since. Re-basing every team to zero at Game 20, when our estimator went live, we are **2nd of 17** at  $+425,074$  over the 80 Games since. At the conservative Game 26 anchor we are still 2nd, leading error404 ai by  $16,471$  — and our per-Game rate is marginally *higher* from 26 than from 20, so the earlier anchor is not the flattering one it looks like.



**Figure 1:** Every team re-based to zero at Game 20, when our estimator went live. Our cumulative net at Game 20 was –354,171, so the whole gap to the leaders is that hole rather than the play since. At the conservative Game 26 anchor we are 3rd, with the top three inside 9,702.



**Figure 2:** The same record un-rebased: cumulative net over all settled Games, all 17 teams, we are the heavy line. 17th of 17 after nine Games. The deficit is Games 1–25, before the estimator was working; the slope after it is the method. Games 82–89 ran with the model channel returning 401 on every call.

## Why we think we succeeded — and where we did not

### Succeeded

- **We built the measuring instrument before the strategy.** Equation (4) recovers  $t$  exactly, so every later decision was settled by replaying it against the real Field rather than argued. Three claims we had written down as fact were falsified within a day.
- **Last place to 5<sup>th</sup>.** 17<sup>th</sup> of 17 after nine Games; 5<sup>th</sup> from Game 76. Second in the field by rate over the last twenty Games; third of seventeen over the 74 Games since the estimator was working.
- **The distribution is tight, and that is the result we would defend.** Over the last thirty Games our mean-to- $\sigma$  is **0.75, second in the field**, on the second-lowest  $\sigma$  of any team; we have three losing Games, second fewest, and our deepest hole is –3,941 while seven teams carry single Games worse than –80,000 (the one team with fewer losing Games is the one ahead of us on the rate).

Over the last *twenty* we are 4th on the same measure at 0.65 — a narrower window that dropped our two best Games. In an insurance book the narrow distribution is the number that matters.

- **It degraded instead of failing.** Eight consecutive 3 $\times$ -weighted Games ran with the model channel completely dark and still banked +254,092, because no scored number ever depended on a model being reachable.
- **We priced uptime correctly.** Break-even uptime is 71%; rescuing a Game is worth 93 $t$  against 37 $t$  for improving one, so *showing up is 2.5 $\times$  being right*. A blind floor posts at T+0 before the Case is decrypted, submissions merge per Line Item, and a supervisor restarts on any crash. Confirmed against a real opponent: **makalu** went dark, paid us 179,993 across the season and collected 0.00 from anyone.

## Did not succeed

- **We spent the first quarter of the tournament building the instrument instead of scoring.** Games 1–25 cost  $-322,595$  — more than our entire current season net. The rate says the decision was right; the total says we paid full price for it.
- **The estimate, not the strategy, is the whole remaining gap — and we learned that late.**  $a = b = t$  reaches 100.3% of optimal play;  $a = b = \hat{t}$  scores  $-50,140$ . The 2,498,118 between them is pure estimation error. We tuned decision rules to their measured argmax and it barely mattered; arguably we worked one layer too low.
- **We diagnosed our largest open lever and deliberately did not ship it.** Policy clause 7.1.5 has two halves and the model reads only the first. Game 74 returned coverage 0.01–0.05 on 20 of 31 Line Items, for 87% of that Game’s penalties. Perfect coverage is worth  $+1,173$ /Game. It is a prompt change that could not be validated without spending the live runner’s quota, six Games before the  $3\times$  window. We think the call was right and the outcome is still a loss.
- **Eight triple-weighted Games ran with the model dark and we did not notice.** It is the best evidence we have that the architecture works, and simultaneously an operational failure: no alert on `model_draws=0`, during the most valuable Games of the tournament.
- **We were beaten on the thing we understood best.** We had written down that the Charge is a cliff and still sat above  $t$  too often; a rival took 136,075 off a single Line Item by charging 22% lower than us. Knowing a mechanism and being calibrated to it are different achievements, and we only closed that gap at Game 63.

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**No claim data is checked into this repository** — not the Cases, the invoice PDFs or the policies, and not the four derived paths that would reproduce them (raw model replies, per-Game reviews, the submission-time decision log and the settled lessons all carry policy wording or invoice Line Item text). All four regenerate locally and are git-ignored. Single-clause quotations appear where the argument needs them, as citation rather than as an archive. Every figure in this document is reproducible from a script in the repository.